Seeds of sovereignty: an empirical investigation into income-mediated empowerment pathways under governmental frameworks

Ravi Ranjan and Mohd Yaseen Khan*

Faculty of Management,
Vivekananda Global University,
Jaipur, Rajasthan, 303012, India
Email: raviranjan.fruitfal@gmail.com
Email: drmykhan76@gmail.com
*Corresponding author

Subhranshu Mohanty

Department of Master of Computer Applications,
Srusti Academy of Management and Technology (Autonomous),
Bhubaneswar, Odisha, 751024, India
Email: drsubhranshumohanty@gmail.com

Abstract: This research examines the impact of government schemes on farmer empowerment and their income enhancement in India. A cross-sectional research design was adopted, and primary data were collected from 471 farmers in Bihar using a structured survey with 37 items, adapted and modified from past studies. Therefore, the objective of the study is to investigate the impact of government schemes and income enhancement on farmer empowerment. Further, it also examines the mediating effect of income enhancement between government schemes and empowerment, and the moderating effect of farm size in these relationships. PLS-SEM is used to examine and validate the hypotheses. The findings show that constructs financial assistance, accessibility and outreach, and scheme focus significantly influence government schemes, whereas farm productivity, wealth accumulation, and government schemes positively influence income enhancement. Additionally, Income Enhancement mediates the link between government schemes and empowerment, while farm size has a positive but insignificant moderating effect.

Keywords: government schemes, farmer empowerment, income enhancement, farm productivity, agricultural development

Reference to this paper should be made as follows: Ranjan, R., Khan, M.Y. and Mohanty, S. (xxxx) 'Seeds of sovereignty: an empirical investigation into income-mediated empowerment pathways under governmental frameworks', *Int. J. Productivity and Quality Management*, Vol. X, No. Y, pp.xxx–xxx.

Biographical notes: Ravi Ranjan is an Assistant Professor in Management with over 10+ years of combined academic and industry experience. Currently associated with NMIET, Pune, he serve as Exam Coordinator, IQAC Coordinator, and Industrial Visit Coordinator for the MBA department. He is pursuing his PhD from Vivekananda Global University, Jaipur and have qualified UGC NET 2018 in management. His expertise includes finance and marketing, along with strong skills in teaching, research, academic administration, and student mentoring. He has contributed to multiple research publications, conferences, FDPs, and academic events, including national seminars funded by ICSSR. He is committed to continuous learning, innovation in education, and contributing effectively to institutional and student development.

Mohd. Yaseen Khan is an Academician having expertise in Marketing and International Business. He has more than 22 years of academic and industry experience and supervised PhD scholars from countries like Bangladesh and Afghanistan, he is a PhD from UTU, Dehradun, India, on the topic – Customised Communication, Value Proposition and Brand Loyalty in Retailing – a study on SERVPERF-RSQS Model. He has been awarded by the President of India for social service in the Scout and Guide and Red Cross Society. He has 15 national/international publications to his credit, along with two book publications. Currently, he is serving as Associate Professor in the Department of Management Studies – Vivekananda Global University, Jaipur-India.

Subhranshu Mohanty is an academician with expertise in Computer Science Engineering. He has more than 23 years of academic experience in reputed institutions of India. He has completed PhD (CSE) from Siksha 'O' Anusandhan Deemed to be University, Bhubaneswar, Odisha. He has published eleven research articles in various National and International journals. He has also published a book titled '*Security in Design Phase of SDLC for Web Applications*'. Currently, he is serving as Associate Professor and Dean-MCA at Srusti Academy of Management and Technology (Autonomous), Bhubaneswar, Odisha.

1 Introduction

Agriculture remains the backbone of the Indian rural economy, with millions of workers depending on it for economic security and social well-being in almost all Indian states. While rapid industrialisation and urbanisation are part of the broader economic picture, the Indian economy still has millions of farmers who depend on the agricultural sector for their livelihoods (Kumar et al., 2018; Amin et al., 2026b; Kushwaha et al., 2026). In recognition of these realities, the Government of India has introduced a number of targeted schemes related to basic aspects of better incomes, vulnerability, and empowerment, through the expansion of credit, the transfer of funds, and the development of technology, improved infrastructure, and direct support to farmers' incomes (Ling and Wahab, 2020). Some examples of these schemes include the Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) program, which provides direct income support payments to eligible farmers, and the AIF, which provides incentives for investments in post-harvest infrastructure and farm mechanisation. Therefore, this paper will address the issue of links between government policies and farmer incomes and empowerment.

Yet a notable gap in research, due to the prevailing discourse on research and policy-related issues, is that, despite numerous government schemes, evidence of their impact on enhancing farmer empowerment beyond income transfer is somewhat fragmented (Kamble and Wankhade, 2017). It is well known that farmers' monetary incomes are improved, which is a very important step, but effectively turning this into increased agency, greater decision-making capacity, access to markets, and improved socio-economic status is easier said than done (Khan and Arif, 2025; Jose et al., 2026a). There is also a lack of exploration of the mechanisms by which government schemes translate into empowerment: is the effect direct, or is income enhancement a mediating mechanism? Secondly, does farm size, for example, small and marginal farm size vis-a-vis medium/large farm size, alter the scheme-income-empowerment relationship? The inability to answer these questions is a serious obstacle in the formulation of nuanced agricultural and rural development policy in India.

Studies in recent years have tended to focus on aspects of farmer empowerment and the role of institutional support. For example, research examined the market orientation of farm systems and the empowerment of women in Indian agriculture (Malapit et al., 2024; Jose et al., 2026b; Shah et al., 2026). Farmer-producer organisations' institutional features and role in promoting inclusive livelihoods have been the subject of other studies (Kumar and Singh, 2024). Systematic reviews of farmer welfare schemes in India reviewed their impacts and challenges and found mixed results for income, productivity, and empowerment (Danyen and Callychurn, 2015; Sharma and Meena, 2024). These studies examine either income enhancement or empowerment, rather than specifying a mediating or moderating mechanism linking them. Therefore, there is a need to investigate the combined government schemes, income enhancement as a mediator, and farm size as a moderator of farmer empowerment in India.

This study is important because it will offer a framework for bridging these gaps by taking a more comprehensive approach to comprehending government assistance for farmers as a way to empower them through financial gains, while acknowledging the diversity brought about by farm sizes. The study sheds light on the moderating effect of farm size and the mediating effect of income improvement. The study makes recommendations on how to improve implementation, targeting, and scheme design in the Indian context while also building on the literature on empowerment, rural livelihoods, and the assessment of agricultural policy. Accordingly, the paper pursues three objectives: to assess the impact of government schemes on farmer empowerment in India; to evaluate how income increase mediates between government schemes and farmer empowerment; and to investigate the moderating role of farm size in the relationship among government schemes, income increase, and farmer empowerment. The study is aimed at farmers located in selected states of India across the small, marginal, and medium farm sizes, looking into both direct and indirect pathways connecting policy support to empowerment outcomes. A cross-sectional research design was applied for the study, and data have been quantitatively analysed to test the proposed relationships, providing fresh evidence to support policy interventions within the agrarian sector.

The rest of the paper is organised as follows. Section 2 is a detailed review of the literature, including the theoretical background and development of the conceptual framework, and the formulation of hypotheses. Section 3 describes the research methodology, including the research design, data collection procedures, measurement of variables, and analytical techniques used. Section 4 presents the data analysis and empirical results, which include measurement model determination, structural model determination, and hypothesis test results. Section 5 discusses the findings in relation to the existing literature and theoretical perspectives. Section 6 emphasises the study's theoretical and policy implications. Finally, Section 7 concludes the paper by summarising the main findings, acknowledging limitations, and suggesting directions for future research.

2 Literature review

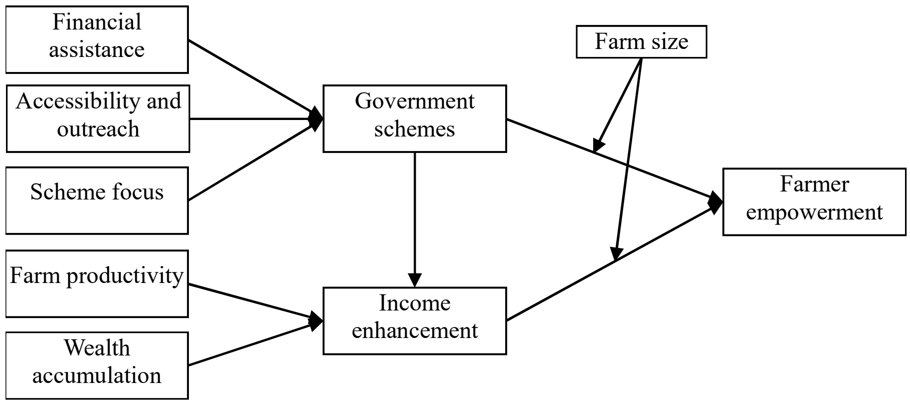
2.1 Theoretical underpinning

The theoretical underpinning of this study is Amartya Sen's Capability Approach (Sen, 1999), which provides a coherent basis for linking the key constructs, i.e., government schemes, income enhancement, and farmer empowerment. Within this framework, government schemes, operationalised through dimensions such as financial assistance, accessibility, and outreach, and scheme focus, serve as institutional mechanisms to increase farmers' access to essential resources, services, and opportunities that enhance their capabilities. These improved abilities are translated into improved economic functioning, with enhanced income, a further function of farm productivity, and the accumulation of wealth. In turn, with the increase in income, farmers strengthen their position in terms of the exercise of choice, reduced dependency, and participation in economic and social decision-making, thus becoming more empowered. Complementing this, Zimmerman's (2000) Empowerment Theory describes how access to resources and improved economic conditions improve self-efficacy and control over livelihood decisions. Additionally, the sustainable livelihoods framework (Chambers and Conway, 1992) provides support for structures by showing how policy interventions affect livelihood assets (financial, physical, and human capital), which are proxied in this research through productivity and wealth accumulation, leading to sustainable income and empowerment outcomes. The inclusion of farm size as a moderating variable reflects heterogeneity in resource endowment structure, which may condition the extent to which capabilities are converted into empowerment. Collectively, these theoretical perspectives establish a clear connection between the study variables and explain how the government interventions discussed translate into empowerment outcomes through the mediating role of income enhancement.

2.2 Conceptual framework

Figure 1 illustrates the study's conceptual framework. The framework highlights the proposed relationships among the main constructs of government schemes, income enhancement, farmer empowerment, and farm size. Government schemes have been conceived as a multidimensional construct of financial assistance, accessibility and outreach, scheme focus, farm productivity, and wealth accumulation. These components cumulatively reflect the coverage and effectiveness of the policy measures adopted by the government with a view to empowering the agricultural economy. The model suggests that government programs directly improve the empowerment of farmers, which consequently means that targeted policy interventions result in higher decision-making capacity, autonomy, and participation of farmers in agricultural markets. Income growth is identified as a mediating variable that supports our assumption that government programs result in farmers' empowerment indirectly, by increasing income stability, productivity, and asset generation. Additionally, farm size is introduced as a moderating variable that affects the strength of the relationship between government programs, income, and farmer empowerment.

Figure 1 Conceptual framework



2.3 Government schemes

To improve the effectiveness and sustainability of government programs in agriculture, financial support is crucial. This includes credit, subsidies, grants, and direct income support. Such backing allows farmers to invest in technology, mechanisation, and quality inputs, which all enhance productivity. The International Food Policy Research Institute (2023) has suggested programs with a coupled design, like PM-KISAN and KCC in India, to offer liquidity and stability in farm income. These programs promote better policy interaction (IFPRI, 2023). As the ministry noted in 2024, evidence shows that initiatives like these not only improve farmers' access to credit but also help them engage with other development programs. This is what makes the policy effective. Moreover, quick access to credit boosts small and marginal farmers' confidence in using new technology and growing their income streams, which increases their overall earnings (Mahajan et al., 2014; Kumar and Singh, 2024). By providing incentives for government initiatives, financial support enhances its perceived effect on rural and agricultural development.

H1 Financial assistance positively impacts government schemes.

The success of government programs in reaching their goals relies on evaluating outreach and accessibility. Beneficiaries like marginal farmers and smallholders gain from effective outreach. According to MANAGE (2023), sharing information through digital outreach, extension services, and local governance helps increase engagement and reduce the risk of exclusion errors. Accessibility is essential for the success of the PM-KISAN, soil health card scheme, and Fasal Bima Yojana in rural India. This includes digital enrollment and last-mile delivery systems. According to MoAFW (2024), improved accessibility helps in timely benefits, swift grievance redressal, and trust in institutional mechanisms. On the contrary, limited outreach or weak institutional networks might create an information asymmetry situation, resulting in low participation and poor performance of the schemes (Kumar and Kushwaha, 2025b; Prasanna and Kushwaha, 2025c). A robust framework for accessibility and outreach is necessary to guarantee that the government's intervention achieves the intended socio-economic and empowerment objectives, according to a study by Kumar and Singh (2023).

H2 Accessibility and outreach positively impact government schemes.

The specific focus and design of government programs shape their success and long-term impact. A clearly defined scheme focuses on farmer needs, regional priorities, and sustainability objectives, giving the needed allocation of funding and consistency in the delivery of policy measures (Johri et al., 2025; Kumar and Kushwaha, 2025a). Schemes designed around improving farm productivity, promoting sustainable practices, and increasing market access better support livelihoods and empower communities (Prasanna and Kushwaha, 2025b). However, fragmented or poorly targeted schemes result in dilution of outcomes and administrative redundancies. Recent analyses have shown that schemes with clear and consistent objectives, such as the AIF and the national mission on sustainable agriculture, increase farmers' income. Additionally, a scheme focus promotes coordination amongst different government programs for cross-sector synergy, including irrigation, credit, and technology adoption (Prasanna and Kushwaha, 2025a; Sharma and Kushwaha, 2025). Therefore, a well-defined scheme focus has a positive impact on participation rates, implementation efficiency, and the overall potential for empowerment resulting from government programs related to agriculture.

H3 Scheme focus positively impacts government schemes.

2.4 *Income enhancement*

Agricultural productivity is the key driver of increasing income in the agricultural sector. Productivity improvement comes from better access to quality inputs, mechanisation, irrigation services, and exploitation of new technologies. Consequently, productivity relates directly to farmers' profitability and economic well-being (Pushparaj et al., 2025; Vijayakumaran et al., 2025). According to the Food and Agriculture Organisation (2023), the National Food Security Mission and the Pradhan Mantri Krishi Sinchayee Yojana have both greatly increased yields and the income levels of farmers in India (Amin et al., 2026a). The Ministry of Agriculture and Farmers Welfare (2024) reported that several peer-reviewed studies also support the notion that higher productivity increases farm revenue while also contributing to a surplus for additional investment in agriculture, or diversification into another agricultural allied sector, such as dairy or horticulture. Reddy and Kumar (2023) identified that implementing systems of sustainable agriculture and precision farming provides greater resource efficiency and enables the longevity of income growth stability. Pandey and Singh (2023) go on to note the urgent need to examine equitable distribution of benefits among farming communities in light of regional differences throughout the country. In the recent past, government support policies, particularly in the case of pulses and oilseeds, have created an enabling environment that has helped enhance farmers' incomes.

H4 Farm productivity positively impacts income enhancement.

Wealth accumulation via ownership of productive assets, savings, and physical capital has a central place in enhancing the income of farming households. By increasing production capacity and resilience to external shocks, asset accumulation aids Indian farmers in sustaining a consistent income (World Bank, 2023). Government programs for infrastructure development, crop insurance, and rural credit help farmers build valuable assets that provide long-term benefits (Kumar and Singh, 2024). Wealthier farmers are also seen as more capable of adopting new practices and diversifying their income into non-agricultural areas. This diversification boosts their stability and financial independence (Deveshwar, 2023). Furthermore, building wealth creates a positive cycle of investment and income growth, serving as collateral for accessing institutional financing (Sharma and Meena, 2024). Therefore, wealth accumulation is a determinant of future income growth rather than merely a reflection of past income and becomes an important intervening variable between government efforts and farmer economic advancement.

H5 Wealth accumulation positively impacts income enhancement.

Government schemes play an important role in promoting farmers' income through the elimination of structural bottlenecks in access to financial, infrastructural, and market resources. Schemes such as PM-KISAN, the Agriculture Infrastructure Fund (AIF), and the Kisan Credit Card (KCC) cover both direct income support and possibilities of productive investment (Rubel et al., 2025; Suryavanshi et al., 2025). These mechanisms aim to stabilise farm produce prices, reduce input costs, and promote technology adoption, all of which will increase income levels, according to IFPRI (2023). Research indicates that a combination of income-transfer initiatives and infrastructure-based interventions tends to boost farm profitability and diversify livelihoods (MoAFW, 2024). In particular, several recent studies indicate that farmers who have engaged to a great extent in multiple government schemes are expected to experience a greater increase in income than farmers who have limited access to government programs (Amin et al., 2025). The accumulated studies suggest that government programs that are effectively targeted and well delivered are helping to enhance income, especially to improve the basis for both farmer empowerment and rural economic resilience in India.

H6 Government schemes positively impacts income enhancement.

2.5 Farmer empowerment

Through capacity agreements, resource access, and formal institution support, government programs indirectly empower farmers. Increased decision-making skills, resource management, knowledge acquisition, and involvement in market and policy systems are essential elements of farmer empowerment in agriculture (Singh and Mehta, 2023). A hybrid effect that arose following programs such as National Mission for Sustainable Agriculture (NMSA), Pradhan Mantri Fasal Bima Yojana (PMFBY), and sustainable agriculture in the highlands of the Himalaya (SHCS) can enable farmers to increase their financial capacity and technical ability, resulting in informed and independent decision-making (Ministry of Agriculture and Farmers Welfare, 2024). NITI Aayog (2023) found that some of these mechanisms furthered digital inclusion, support services, and training that would lower farmers' financial pressure and enhance their confidence. Indeed, multiple studies cite how increased access to government support further stimulates participation connected to social and economic empowerment with producer organisations and cooperatives and self-help groups (SHGs) (Patel and Rani, 2023). Additionally, programs providing financial literacy and gender sensitive programs provided further awareness for empowerment specifically with SHGs and other marginalised or rural communities (Chatterjee et al., 2023). Thus, the proper mechanisms for mobilising empowerment, whether individual or collective is facilitated through government support mechanisms to assist farmers agencies and economic resilience.

H7 Government schemes positively impacts farmer empowerment.

Consequently, income improvement is an important factor related to farmer empowerment with respect to improved access, reduced reliance on intermediaries, and traditional sources for credit and inputs, and an enhanced capacity to make choices. Higher incomes enable farmers to invest in productive assets, education, and technologies; cumulatively, these factors contribute to increasing farmers control over their own livelihood strategies. Studies have demonstrated that higher income provides greater confidence and promotes increased engagement with local governance based on livelihood terms of local economy (World Bank, 2023). In India, programs such as PM-KISAN, e-NAM (national agriculture market) and rural infrastructure development fund (RIDF) promote food security and have positive relationships with empowerment outcomes. Income growth boosts social status and financial independence, particularly for women farmers who feel empowered in household decision-making and have reduced financial vulnerability. There is evidence that stable income, in addition to the economic aspects, supports empowerment through sustainability in agricultural practices, which assists in helping farmers learn and experiment with new ideas and practices, and through an element of psychological safety (Narayanan and Bhatia, 2024). Income enhancements, therefore, improve financial well-being, but they are also an important source of empowerment across the economic, social, and personal spheres.

H8 Income enhancement positively impacts farmer empowerment.

3 Research methodology

This study is a descriptive and cross-sectional study. The primary data was collected through a quantitative data collection method using a structured questionnaire with farmers in Bihar, between the months of July and October 2025. All farmers who received government agricultural programs or were aware of these government initiatives were the study population. Farmers were selected for the study using a purposive sampling technique. This ensures that there was adequate knowledge and first-hand experience of government agricultural schemes by respondents, which forms the crux of the research. Given the nature of the constructs that are under investigation, such as government schemes, income enhancement, and the empowerment of farmers, it was crucial to select persons who were either beneficiaries of such schemes or were aware enough of them. For analyses like partial least squares-structural equation modelling (PLS-SEM), a sample size of 471 farmers was adequate (Hair et al., 2021). The sample size for this study was determined based on the widely accepted '10-times rule' recommended in PLS-SEM literature. According to the minimum sample size should be at least ten times the maximum number of structural paths directed at any latent construct in the model or the largest number of indicators used to measure a single construct (Hair et al., 2019). Data collection involved one-on-one interviews held at convenient locations for the participants, including community centres, agriculture offices, and some farmers' homes. Confidentiality and trust were important parts of the process. Each participant received full information about the study, including its voluntary nature and the confidentiality of their responses. They also signed a written consent form after being informed.

The purpose of the questionnaire was to evaluate eight constructs, including farmer empowerment, income, and government programs. It was modified from previously validated research tools. 37 items measured on a seven-point Likert scale (1-strongly disagree and 7 being strongly agree) were utilised to assess the eight constructs. Demographic characteristics such as age, education level, size of farm, and experience were included in the first section of the questionnaire, and the constructs were included in the second section of the instrument. A pilot study was conducted to assess the sufficiency and validity of the measurement tool before actual data collection began. The pilot survey was conducted with 68 participants, who represented the target population. Reliability analysis was conducted using Cronbach's alpha coefficients for each construct. The scores showed good internal consistency, with Cronbach's alpha values exceeding the recommended 0.70 for all variables. These results helped confirm satisfactory reliability across all constructs in the pilot sample. Refinements to the questionnaire were made based on minor feedback regarding item clarity and wording.

Data analysis was performed using SmartPLS 4.0 software in order to test the measurement and structural models. The use of PLS-SEM is suitable for these data because of its appropriateness in dealing with complex models, samples of a moderate size, and non-normality in distributions. Descriptive statistics included means, standard deviations, and frequencies in order to summarise sample characteristics, while inferential statistics measured the relationship among constructs. Reliability and validity were checked through Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE).

4 Data analysis

4.1 Respondent's profile

Table 1 presents the demographic characteristics of the 471 respondents that participated in the study. The results indicate that most of the participants were male (73%), and only 27% were female, a factor that indicates farming in the study region is still a male-dominated field. Regarding age, 36% of the respondents fell under the age group 31–40 years, 28% between 18–30 years, 21% between 41–50 years, and 15% between 51–60 years. In terms of marital status, 34% of people were single and 66% of people were married.

56% of the population had completed high school, 32% had completed college, and 12% had completed postgraduate studies. These percentages indicate a moderate level of educational attainment. In farming experience, 32% had 3–5 years, 29% had 6–8 years, and 22% had more than 8 years of farming involvement. It is seen from the data that a majority of the respondents have ample practical experience and are thereby capable of giving informed insight into how government schemes work to affect their incomes and empower them. Overall, the profile indicates a diversified sample representing a variation in demographic, educational, and experiential backgrounds among Bihar farmers.

Table 1 Respondent's profile

Gender	*f*	%	Age	*f*	%
Male	344	73	18–30 years	132	28
Female	127	27	31–40 years	169	36
			41–50 years	99	21
Marital status	*f*	%	51–60 years	71	15
Married	311	66			
Single	160	34	*Involve in farming*	*f*	%
Educational qualifications	*f*	%	>3 yrs	80	17
High school	264	56	3–5 yrs	151	32
Undergraduate	151	32	6–8 yrs	136	29
Postgraduate	56	12	Above 8 yrs	104	22

Table 2 Assessment of scale items

Variables	*Items*	*Outer loading*	*VIF*	*Variables*	*Items*	*Outer loading*	*VIF*
Accessibility and outreach	AO1	0.884	3.024	Government schemes	GS1	0.883	3.292
	AO2	0.919	3.783		GS2	0.887	3.517
	AO3	0.891	2.657		GS3	0.914	4.069
	AO4	0.792	1.782		GS4	0.752	1.900
Financial assistance	FA1	0.752	1.712		GS5	0.785	1.739
	FA2	0.905	3.378	Income enhancement	IE1	0.801	1.858
	FA3	0.803	2.200		IE2	0.840	2.484
	FA4	0.866	2.790		IE3	0.883	3.096
	FA5	0.877	2.760		IE4	0.880	3.783
Farmer empowerment	FE1	0.902	3.476		IE5	0.779	2.616
	FE2	0.873	2.953	Scheme focus	SF1	0.799	1.247
	FE3	0.831	2.634		SF2	0.782	2.249
	FE4	0.864	2.824		SF3	0.741	1.465
	FE5	0.834	2.419		SF4	0.751	2.471
Farm productivity	FP1	0.757	1.705	Wealth accumulation	WA1	0.916	3.717
	FP2	0.890	3.118		WA2	0.940	4.786
	FP3	0.813	2.252		WA3	0.916	3.626
	FP4	0.865	2.672		WA4	0.906	3.361
	FP5	0.781	1.851				

4.2 Measurement items assessment

Table 2 presents the assessment of the measurement items used to operationalise the study's constructs. Consistent with the recommended criterion that outer loadings should exceed 0.70 to indicate adequate indicator reliability (Sarstedt et al., 2017), the majority of items in this study meet this threshold, demonstrating strong individual contributions to their respective constructs. To further examine multicollinearity among indicators, VIF values were evaluated. Although a few items show VIF values slightly above the conservative guideline of 3 (e.g., 2.657, 2.790, 2.760, 2.953), all values remain well below the more relaxed upper limit of 5, which is commonly accepted in PLS-SEM literature (Sarstedt et al., 2014). These results confirm that multicollinearity is not a concern, and the scale items exhibit acceptable psychometric properties for inclusion in the structural model.

4.3 Quality criteria assessment

Table 3 shows the reliability and convergent validity of the variables in this study. Cronbach's alpha ($\\alpha$) was used to assess the internal consistency suitability of each variable's scale items. As a general rule, $\\alpha$ should be above 0.70; therefore, all of the variables in this study had values above 0.70, confirming the reliability of the measurement scales, with the lowest value being 0.757. In addition, for reliability and convergent validity, the study used CR (rho_A and rho_C) and AVE, respectively. All the values of CR rho_A and CR rho_C range between 0.70 and 0.95, which is acceptable, and thus, construct reliability is satisfactory. However, for convergent validity, not all variables reached the threshold value of 0.50 for AVE as set by Hair and Alamer (2022). Although most variables had AVE values greater than 0.50, constructs such as 'farm productivity' and 'scheme focus' registered less than 0.50, at 0.677 and 0.554, respectively. As such, convergent validity is not established across the board for the constructs, though the values of CR are acceptable, which could imply a trade-off between the construct's explanatory power, represented by AVE, and the internal consistency represented by CR. This is usually considered acceptable if the CR values are high, according to Hair et al. (2019).

Table 3 Construct reliability and validity

Variables	α	*CR rho_A*	*CR rho_C*	*AVE*
Accessibility and outreach	0.895	0.904	0.927	0.762
Farm productivity	0.879	0.884	0.913	0.677
Farmer empowerment	0.913	0.919	0.935	0.742
Financial assistance	0.897	0.903	0.924	0.710
Government schemes	0.899	0.904	0.926	0.716
Income enhancement	0.893	0.898	0.921	0.702
Scheme focus	0.757	0.833	0.832	0.554
Wealth accumulation	0.939	0.940	0.956	0.846

Table 4 gives the HTMT ratio of correlations between all variables in this study. It is applied as a test for discriminant validity. According to the criteria, the values of HTMT must fall below 0.85. However, as suggested by Henseler et al. (2015), values as high as 0.90 may be acceptable. The HTMT ratios in this study fall between the lowest value represented here but not actually shown and a maximum value of 0.895. Since the highest observed HTMT value falls below the threshold of 0.90, the analysis indicates that the constructs are distinct from each other. Therefore, there is adequate discriminant validity among the reflective constructs in this study based on Hair and Alamer (2022).

Table 4 Heterotrait-Monotrait (HTMT) ratio of correlations

	Accessibility and outreach	*Farm productivity*	*Farmer empowerment*	*Financial assistance*	*Government schemes*	*Income enhancement*	*Scheme focus*	*Wealth accumulation*
Accessibility and outreach								
Farm productivity	0.704							
Farmer empowerment	0.670	0.747						
Financial assistance	0.733	0.799	0.746					
Government schemes	0.754	0.764	0.812	0.770				
Income enhancement	0.725	0.858	0.749	0.861	0.887			
Scheme focus	0.812	0.641	0.589	0.684	0.733	0.639		
Wealth accumulation	0.793	0.798	0.859	0.806	0.856	0.634	0.700	

4.4 Structural equation model

Figure 2 path relationship diagram showing the structural model of the significant and non-significant predictive relationships amongst the study constructs. The path analysis makes it clear that the highest significant predictors of income enhancement $R^2 = 0.803$) are wealth accumulation $\beta = 0.486$, $p < 0.001$) and financial assistance $\beta = 0.330$, $p < 0.001$). Farm productivity $\beta = 0.227$, $p < 0.001$) also contributes significantly towards income enhancement. Further, government schemes $R^2 = 0.609$) are significantly explained by financial assistance $\beta = 0.263$, $p < 0.001$), accessibility and outreach $\beta = 0.263$, $p < 0.001$), and scheme focus $\beta = 0.289$, $p < 0.001$). Most importantly, there exists a strong and significant path from income enhancement to farmer empowerment $\beta = 0.741$, $p < 0.001$) that posits the improvement of income as the key to farmer empowerment; $R^2 = 0.769$. Last but not least, government schemes had a non-significant impact on farm size ($\beta = 0.036$, $p = 0.431$) and a significant but weak direct effect on farmer empowerment ($\beta = 0.111$, $p = 0.007$). According to the entire model, targeted programs, accessibility, and government intervention all have an impact on how successful the schemes are perceived to be. This affects the increase in income and, eventually, the empowerment of farmers.

Figure 2 Path relationship diagram (see online version for colours)

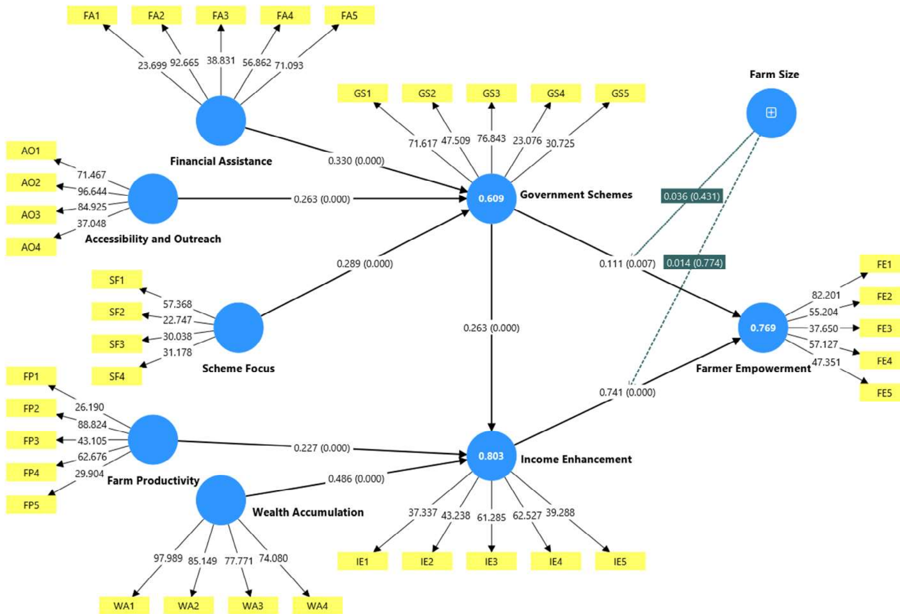


Table 5 Hypothesis testing using bootstrapping

Hypotheses		*β*	*Mean*	*STDEV*	*T stat.*	*P values*	*Decision*
H1	Financial assistance -> government schemes	0.330	0.330	0.043	7.594	0.000	Supported
H2	Accessibility and outreach -> government schemes	0.263	0.264	0.053	4.981	0.000	Supported
H3	Scheme focus -> government schemes	0.289	0.290	0.043	6.782	0.000	Supported
H4	Farm productivity -> income enhancement	0.227	0.227	0.033	6.866	0.000	Supported
H5	Wealth accumulation -> income enhancement	0.486	0.486	0.042	11.673	0.000	Supported
H6	Government schemes -> income enhancement	0.263	0.263	0.044	5.988	0.000	Supported
H7	Government schemes -> farmer empowerment	0.111	0.114	0.041	2.680	0.007	Supported
H8	Income enhancement -> farmer empowerment	0.741	0.739	0.037	19.852	0.000	Supported

Table 5 shows the results of the hypothesis testing by bootstrapping, validating all eight proposed hypotheses. The analysis of the structural model reveals that Income Enhancement is the strongest predictor in the model, as evidenced by the highly significant path from income enhancement to farmer empowerment (H8), with the largest path coefficient (β = 0.741) and highest T-statistic (19.852). The second-strongest effect is the one of wealth accumulation on income enhancement (H5) with β = 0.486 and T-stat = 11.673. All three independent variables, namely financial assistance (H1), accessibility and outreach (H2), and scheme focus (H3), significantly and positively predict government schemes, with respective β values of 0.330, 0.263, and 0.289 (all $p < 0.001$). Moreover, although government Schemes has a significant effect on both income enhancement (H6) with β = 0.263 ($p < 0.001$) and farmer empowerment (H7) with β = 0.111 (p = 0.007), this effect is much weaker as compared to the impact mediated through income enhancement (H8). These findings cumulatively establish a strong sequential link from government Scheme drivers to farmer empowerment, primarily through income improvement.

Table 6 Indirect effects

Hypotheses	*β*	*Mean*	*STDEV*	*T stat.*	*P values*	*Decision*
			Mediating effects			
Government schemes -> income enhancement -> farmer empowerment	0.195	0.194	0.031	6.213	0	Supported
			Moderating effects			
Farm size -> farmer empowerment	0.032	0.032	0.033	0.986	0.324	Not supported
Farm size x government schemes -> farmer empowerment	0.036	0.037	0.046	0.788	0.431	Not supported
Farm size x income enhancement -> farmer empowerment	0.014	0.012	0.048	0.288	0.774	Not supported

Table 6 summarises the results of the mediating and moderating effects for the structural model. In terms of mediation, it can be found that the hypothesis concerning the indirect effect of Income enhancement between government schemes and farmer empowerment is supported with a path coefficient of β = 0.195, which is significant (T-stat = 6.213, $p < 0.001$). That means the positive influence of government schemes on farmer empowerment is partially channeled and significantly magnified through the intermediate construct of income enhancement. In contrast, concerning the moderating effect analysis of Farm Size, it could be found that all three proposed hypotheses about moderation are not supported. Specifically, the effects of farm size on empowerment (β = 0.032, p = 0.324), farm size × government schemes on empowerment (β = 0.036, p = 0.431), and farm size × income enhancement on empowerment (β = 0.014, p = 0.774) are all statistically not significant. Therefore, the size of a farmer's landholding has no discernible impact on the direct or interactive relationships that drive farmer empowerment, even though income enhancement is a key component in this model.

5 Discussion

Currently, the conclusion that the financial assistance, accessibility and outreach, and scheme focus variables have a meaningful and positive influence on the overall effectiveness of government programs closely resembles previous empirical work, which pointed to the need for intentional institutional and financial support for agricultural development. For example, many successful government programs incorporate monetary subsidies, assistance with infrastructure, and have an accessible means of communication to access and influence smallholder farmers (Chand et al., 2021). Financial assistance, in particular with regard to access to credit, crop insurance, and income support, has been noted to enhance farmers' capacity to assume risk and adopt new technologies (Rao and Tripathi, 2023). The availability of outreach and accessibility through digital platforms, extension services, and local government enters the rural-urban information gap (Kumar and Rani, 2022). The focus and design of the schemes on specific agricultural challenges, such as climate resilience, input cost reduction, and sustainable productivity, further enhance policy effectiveness (Bisht and Singh, 2024). Therefore, the present study reinforces the argument that alignment of the financial, informational, and structural dimensions underlines the success of government intervention in rural and agricultural contexts.

The finding that agricultural performance, capital accumulation, and institutional support are complementary mechanisms to increase income levels in rural areas is supported by the finding that farm productivity, wealth accumulation, and government Initiatives positively influence income enhancement. Improved farm productivity, like the availability of higher-quality inputs alongside irrigation and technology, positively influences profitability and ultimately livelihoods (Kumar et al., 2023). Additionally, the wealth accumulation safety net established through savings, land, and diversification into other agricultural enterprises further enables reinvestment in the farm sector (Gupta and Jha, 2021). In addition to the accumulation of wealth through production, government programs, such as the soil health card scheme and PM-KISAN, assisted smallholders through the provision of knowledge resources and liquidity, which in turn produced higher income growth (Narayan et al., 2024). These findings strengthen the development economist's position that agrarian economies' poverty cycle can be broken through the synergies of productivity, policy incentives, and accumulation of household assets (Sarkar and Sinha, 2022). This study thus expounds the interplay of institutional support and personal economic capacity in realising effective and long-term income enhancement for Indian farmers.

The idea that empowerment presents multifaceted constructs, namely economic, informational, and social, is also further supported in that both government programs and income enhancement contribute to farmer empowerment. In general, I believe that government initiatives that give farmers access to local markets, financial stability, and capacity-building programs will gradually increase farmer agency and confidence in their choices (Bhattacharya and Sharma, 2022). Because it gives farmers more purchasing power, reduces their reliance on unofficial lending markets, and promotes better farming methods, income enhancement would also result in greater empowerment (Joshi and Meena, 2023). According to Singh and Yadav (2024), state-led agricultural initiatives and financial inclusion also boost farmers' self-efficacy and collective bargaining power. Hence, the present study adds to the existing literature by empirically establishing that institutional and economic enablers work in tandem to promote farmer empowerment in India's rural landscape.

The mediation effect of income increase with respect to government program and farmer empowerment refers to the manner in which government interventions facilitate the process of empowerment. In essence, the economic dimension exists because the increased incomes for farmers from successful government schemes support agency and participation in decision-making, even if prior studies have demonstrated a direct connection between government policy programs and farmer empowerment (Patel and Thomas, 2021). Income is a facilitator or an outcome of success, and also leads to the availability of education, technology, and credit, while also having as much of an impact on enabling farmers to actively participate in their communities and agricultural development (Kumar and Reddy, 2024). This also constitutes a confirmation of what the sustainable livelihoods framework maintains-income diversification and financial stability lie at the core of empowerment (Chowdhury and Dey, 2022). Therefore, the findings of the study are theoretically valuable; they confirm that economic empowerment acts as a mediating link between policy intervention and socio-political empowerment, thus identifying the integrative role of income within rural transformation.

The findings, which indicate a positive but negligible moderating effect of farm size between government schemes and farmer empowerment, imply that government programs are generally inclusive and advantageous to farmers of all land holding types. Previous research has shown conflicting data: larger farms have better access to information and inputs, while smallholders frequently benefit more proportionately from welfare programs that target them more specifically (Das and Pradhan, 2021). Additionally, the absence of moderation may point to a better plan's design and execution, wherein financial aid and training initiatives are given based on need rather than farm size (Suresh and Kaur, 2023). Besides, the growing focus on inclusive agricultural policies and digital outreach has minimised structural gaps among farmers (Roy and Pal, 2024). This finding shows that government programs aimed at empowerment impact more than just land ownership inequalities. They also support the goal of inclusive rural development.

Therefore, it would seem that the empowering effect of income is relatively consistent across landholding size systems, judging from the positive but very slight moderating influence of income increase on farmer empowerment. This result corroborates the study by Singh and Patel (2023) that stated the increase in income, rather than the size of the farm, was the real factor that propelled the Indian farmer in his journey toward empowerment. Minimal moderation may suggest that subjective factors like knowledge, confidence, and social capital may be even more important in empowering people than structural factors like land size. Therefore, this may support a more contemporary definition of empowerment in the agricultural context, which takes into consideration evidence of both psychological and economic transformation and does not depend upon farm size alone.

6 Implications

6.1 Theoretical implications

This study presents multiple potential theoretical contributions to the existing academic scholarship concerning agricultural development, rural empowerment, and public policy. First, it enhances the understanding of interrelated mechanisms that connect government action, income increase, and empowerment in the Indian agricultural context. Empirically validating the mediating role of income enhancement in the relationship between government schemes and farmer empowerment. By showing that financial gains are a crucial conduit through which institutional interventions result in real empowerment, this study expands on the sustainable livelihoods framework and the empowerment theory (Zimmerman, 2000). Secondly, this study provided support for the theoretical claim that program design and inclusiveness are the determining features of policy success – it showed that financial assistance, accessibility, and program scope were all important predictors of perceived effectiveness for government programs (Bisht and Singh, 2024). There is some empirical support for Sen's (1999) inclusive development approach because the lack of significance of the moderating effect of farm size also suggests that the outcomes of empowerment and income enhancement are not structurally limited by variation in landholdings. This research thereby contributes to the existing models of agricultural empowerment by integrating the institutional, economic, and socio-structural dimensions into one coherent analytical framework.

6.2 Managerial (policy) implications

These findings carry important managerial and policy implications, not only for legislators but also for practitioners of rural development and financial institutions. The implications underline the importance of considering well-planned and focused government programs that address financial inclusion, information accessibility, and localised engagement all at once. For instance, it is abundantly clear that creating digital channels and decentralised systems of governance necessitates enhancing the schemes' accessibility and transparency in order to assist the poorest farmers in obtaining efficient subsidy, credit, and training programs. A multifaceted result can be obtained by combining financial assistance, capacity building, and technology appropriation. Additionally, there is a positive correlation between certain government programs and increases in farmer agency and income. Additionally, policymakers can successfully pursue agricultural agendas for inclusive farmer program delivery that prioritise equitable outcomes without any measure of scale differentiation because farm size is nearly irrelevant. The creation of a monitoring and assessment system that includes more than just output indicators should be the main goal of development organisations. Last but not least, cooperation between agritech companies, rural banks, and governments will increase each party's delivery efficiency and create the framework for a longer-term plan to enable agriculture as a key driver of the transformation of Indian agricultural social infrastructure.

7 Conclusions

The intricate relationships between Indian government schemes, income enhancement, and farmer empowerment are empirically supported by this study. Three key elements that contribute to the success of government programs are financial assistance, accessibility and outreach, and scheme focus. Furthermore, various combinations of wealth, productivity, and government programs mediate income enhancement. Similarly, it has been discovered that government programs and income enhancement directly and favourably affect farmer empowerment. More significantly, the relationship between government programs and farmer empowerment is mediated by income enhancement, indicating that policy interventions that promote economic enhancement are essential to attaining the social and psychological components of empowerment and, eventually, the objective of raising rural livelihoods. Even out of the large variation in farm size what little positive effect is seen these results present is that which supports the inclusion in Indian agricultural programs of large and small farmers' issues. When we look at the total picture of what we see in our research, large and small-scale farmers do indeed benefit from empowerment. Also, we see through this study, which looked at the economic and social institutions' interaction empirically, and which in turn brought to light policy recommendations that promote income and support empowerment sustainably. We put forth a case to take forward theoretical and practical discussion on the topic of farmer empowerment.

The study is not without certain limitations that suggest avenues for future studies. First, the cross-sectional nature of the study limits the ability to establish causal inferences among government schemes, income enhancement, and farmer power in the country; thus, future studies could opt for longitudinal or panel data approaches to capture dynamic changes over time. Second, the study is limited to the geographical area of farmers in Bihar and may therefore have limited generalisability to varying agro-economic and socio-cultural settings in India; therefore, comparative studies across various states or regions are recommended. Third, the use of self-reported survey data may introduce common method bias and subjective interpretation; therefore, mixed-method approaches or secondary data may be considered for inclusion in future research to enhance robustness. Additionally, although farm size is included as a moderator in the study, other factors that may be influential, such as gender, digital literacy, access to markets, social capital, and climate vulnerability, are not included, and this could be a subject of future research to further enrich the model. Moreover, future studies could examine higher-order constructs or use multiple-group analysis within PLS-SEM to assess heterogeneity across farmer categories. Finally, experimental or policy evaluation studies could yield more in-depth information about the effectiveness of government interventions, thereby making more actionable and context-specific policy recommendations

Declarations

We do not have a conflict of interest.

This study includes human participation. Participants were informed about the study's objectives, and their answers were treated with strict confidentiality and anonymity. Their written consent was collected.

References

Amin, R., Kushwaha, B. P., Hannan, A., Islam, S., Haque, M. and Tushar, H. (2025) 'Technological foundations and structural dynamics of supply chain networks in Bangladesh's Ready-Made Garment Industry', in Carvalho, L., Reis, L., Figueiredo, A.N.d.A.B. and Russo, N. (Eds.): *Leveraging Technology for Organizational Adaptability*, pp.323–346, IGI Global Scientific Publishing. Pennsylvania, USA.

Amin, R., Kaur, G., Kushwaha, B.P., Lamsal, R., Yajid, M.S.A. and Jagtap, J. (2026a) 'Analysing the technological foundations shaping the structural features of supply chain networks in the Bangladeshi ready-made garment industry', *International Journal of Process Management and Benchmarking*, Vol. 22, No. 3, pp.340–360.

Amin, R., Kushwaha, B.P., Muppani, V.R., Sairam, P.S. and Siddhey, P.K. (2026b) 'Navigating sustainability in supply chain management: assessing the role of corporate communication and sustainable supply chain practices', *Corporate Social Responsibility and Environmental Management*, pp.1–18.

Bhattacharya, R. and Sharma, N. (2022) 'Public interventions and farmer empowerment in rural India: a multidimensional approach', *Journal of Rural Studies*, Vol. 94, pp.102–113.

Bisht, R. and Singh, V. (2024) 'Evaluating the impact of agricultural policy design on sustainable productivity in India', *Agricultural Economics Review*, Vol. 45, No. 2, pp.134–150.

Bland, J.M. and Altman, D.G. (1997) 'Statistics notes: Cronbach's alpha', *BMJ*, Vol. 314, No. 7080, p.572.

Chambers, R. and Conway, G. (1992) 'Sustainable rural livelihoods: practical concepts for the 21st century' [online] https://opendocs.ids.ac.uk/articles/report/Sustainable_rural_livelihoods_practical_concepts_for_the_21st_century/26473510/1?file=48232906 (accessed 2 February 2026).

Chand, R., Kumar, S. and Pandey, A. (2021) 'Financial inclusion and agricultural growth: the role of policy interventions in India', *Indian Journal of Agricultural Economics*, Vol. 76, No. 3, pp.301–318.

Chatterjee, R., Gupta, M. and Patel, D. (2023) 'Government interventions and women's empowerment in agriculture: evidence from rural India', *Journal of Rural Studies*, Vol. 98, No. 4, pp.112–124.

Chowdhury, A. and Dey, S. (2022) 'Income diversification and empowerment: revisiting the sustainable livelihoods framework in rural India', *Development Studies Review*, Vol. 40, No. 4, pp.556–570.

Danyen, S. and Callychurn, D.S. (2015) 'Total quality management success factors and their relationships with performance measures in the food industry: a Mauritian case study', *International Journal of Productivity and Quality Management*, Vol. 16, No. 3, pp.249–266.

Das, S. and Pradhan, R. (2021) 'Landholding and policy outcomes: assessing inclusivity of agricultural welfare schemes in India', *Economic and Political Weekly*, Vol. 56, No. 39, pp.47–56.

Deveshwar, A. (2023) 'Overview of agricultural subsidies in India and its impact on environment and livelihoods', *Current World Environment Journal*, Vol. 18, No. 1, pp.23–35.

Gupta, A. and Jha, S. (2021) 'Wealth accumulation and rural income growth: evidence from Indian farming households', *Journal of Development Economics*, Vol. 153, p.102723.

Hair, J. and Alamer, A. (2022) 'Partial least squares structural equation modeling (PLS-SEM) in second language and education research: guidelines using an applied example', *Research Methods in Applied Linguistics*, Vol. 1, No. 3, p.100027.

Hair, J.F., Hult, G.T.M., Ringle, C.M. and Sarstedt, M. (2021) *A Primer on Partial Least Squares Structural Equation Modeling* (*PLS-SEM*), 3rd ed., SAGE Publications, Thousand Oaks, California.

Hair, J.F., Risher, J.J., Sarstedt, M. and Ringle, C.M. (2019) 'When to use and how to report the results of PLS-SEM', *European Business Review*, Vol. 31, No. 1, pp.2–24.

Henseler, J., Ringle, C.M. and Sarstedt, M. (2015) 'A new criterion for assessing discriminant validity in variance-based structural equation modeling', *Journal of the Academy of Marketing Science*, Vol. 43, pp.115–135.

International Food Policy Research Institute (IFPRI). (2023) *Evaluation of PM-KISAN: Impact on Farm Income and Investments*, Government of India [online] https://desagri.gov.in/wp-content/uploads/2024/04/2022-23-The-Impact-of-PM-KISAN-Scheme-on-the-Farm-Income-of-Beneficiaries-in-Uttar-Pradesh.pdf (accessed 5 March 2026).

Johri, A., Singh, R.K., Kushwaha, B.P., Alhumoudi, H., Alakkas, A. and Khoja, M. (2025) 'Leveraging open innovation for e-commerce success: the contingent role of accounting information systems and artificial intelligence', *Journal of Innovation and Knowledge*, Vol. 10, No. 4, p.100737.

Jose, E.M.K., Kushwaha, B.P., Chaudhary, B.P. and Sharma, A. (2026a) 'Digital transformation that impacts work-life balance and job, life and career satisfaction in female faculty members from the digital era', *International Journal of Business and Systems Research*, Vol. 20, No. 1, pp.70–90.

Jose, E.M.K., Kushwaha, B.P., Chaudhary, B.P. and Singh, V.N. (2026b) 'Integration of emotional intelligence with machine intelligence to enhance internal service performance in the service industry', *International Journal of Applied Management Science*, Vol. 18, No. 1, pp.111–135.

Joshi, R. and Meena, P. (2023) 'Income-led empowerment: a study of Indian farmers under new agricultural reforms', *Asia Pacific Journal of Rural Development*, Vol. 33, No. 1, pp.75–91.

Kamble, R. and Wankhade, L. (2017) 'Perspectives on productivity: identifying attributes influencing productivity in various industrial sectors', *International Journal of Productivity and Quality Management*, Vol. 22, No. 4, pp.536–566.

Khan, M.R. and Arif, M.Z.U. (2025) 'Operational efficiency through disruptive innovation in dairy business: evidence from an emerging economy', *International Journal of Productivity and Quality Management*, Vol. 45, No. 1, pp.66–93.

Kumar, P. and Rani, M. (2022) 'Digital outreach and agricultural inclusion: a study on government extension systems in India', *Information Technology for Development*, Vol. 28, No. 4, pp.897–914.

Kumar, R. and Singh, P. (2024) 'Institutional role of farmer producer organizations (FPOs) in promoting inclusive livelihoods in India', *Indian Policy and Public Review*, Vol. 6, No. 2, pp.45–58.

Kumar, S., Sharma, P. and Rao, V. (2023) 'Farm productivity, technology adoption, and income growth: evidence from rural India', *Agricultural Systems*, Vol. 210, p.103621.

Kumar, U.S., Shrivastava, R.L. and Rajasree, O.P. (2018) 'Quality factors and performance in citrus industry', *International Journal of Productivity and Quality Management*, Vol. 25, No. 4, pp.480–490.

Kumar, V. and Kushwaha, B.P. (2025a) 'A systematic literature review of virtual reality in tourism marketing using a mixed method', *Cogent Business and Management*, Vol. 12, No. 1, p.2528161.

Kumar, V. and Kushwaha, B.P. (2025b) 'Advancing sustainable tourism through virtual reality technology: a systematic literature review using the ADO framework', *Journal of Hospitality and Tourism Insights*, Vol. 8, No. 10, pp.3923–3939.

Kumar, V. and Reddy, R. (2024) 'Economic mediation of empowerment: Insights from government-led agricultural programs in India', *Journal of Agricultural Policy and Development*, Vol. 12, No. 2, pp.45–63.

Kushwaha, B.P., Paul, J. and Koles, B. (2026). 'To share or not to share'? Informational privacy paradox', *International Journal of Consumer Studies*, Vol. 50, No. 2, p.e70182.

Ling, E.K. and Wahab, S.N. (2020) 'Integrity of food supply chain: going beyond food safety and food quality', *International Journal of Productivity and Quality Management*, Vol. 29, No. 2, pp.216–232.

Mahajan, R., Garg, S. and Sharma, P.B. (2014) 'Food safety in India: a case of deli processed food products Ltd', *International Journal of Productivity and Quality Management*, Vol. 14, No. 1, pp.1–20.

Malapit, H.J., Sraboni, E. and Quisumbing, A.R. (2024) 'Market orientation of farming systems and women's empowerment in agriculture: evidence from India', *Food Security*, Vol. 16, No. 2, pp.321–337.

MANAGE (National Institute of Agricultural Extension Management). (2023) *Journal of Agricultural Extension Management*, January–June 2023 Issue [online] https://www.manage.gov.in/publications/journal/jan-jun2023.pdf (accessed 11 November 2025).

Ministry of Agriculture and Farmers Welfare (MoAFW) (2024) *Annual report 2023–24: Enhancing Farmer Income and Empowerment Through Government Initiatives*, Government of India.

Narayan, D., Roy, S. and Dutta, A. (2024) 'Impact of PM-KISAN scheme on rural income dynamics: evidence from Indian states', *Indian Economic Journal*, Vol. 72, No. 1, pp.1–22.

Narayanan, S. and Bhatia, R. (2024) 'Pathways to empowerment: Income, technology adoption, and sustainable agriculture in India', *Agricultural Economics Research Review*, Vol. 37, No. 1, pp.88–103.

NITI Aayog. (2023) 'Empowering farmers through policy interventions: assessment of central government schemes', *Government of India* [online] https://www.niti.gov.in (accessed 11 November 2025).

Pandey, A. and Singh, V. (2023) 'Technological innovation and farm productivity growth: evidence from Indian agriculture', *Journal of Rural Development*, Vol. 42, No. 1, pp.55–70.

Patel, J. and Thomas, L. (2021) 'Government schemes, income growth, and empowerment: a mediational analysis in Indian agriculture', *Journal of Development Policy Studies*, Vol. 8, No. 3, pp.89–104.

Patel, M. and Rani, S. (2023) 'Farmer producer organizations and empowerment: the role of collective action in enhancing resilience', *Indian Journal of Agricultural Economics*, Vol. 78, No. 2, pp.145–162.

Prasanna, A. and Kushwaha, B.P. (2025a) 'Conversational AI chatbots research: unveiling evolutionary trends and thematic clusters using bibliometric analysis', *International Journal of Process Management and Benchmarking*, Vol. 21, No. 2, pp.172–196.

Prasanna, A. and Kushwaha, B.P. (2025b) 'Generative AI in marketing: foundations, trends, and future research propositions', *Human Behavior and Emerging Technologies*, Vol. 2025, No. 1, p.5542513.

Prasanna, A. and Kushwaha, B.P. (2025c) 'Transforming marketing landscapes: a systematic literature review of generative AI using the TCCM model framework', *Management Review Quarterly*, Vol. 76, No. 1, pp.421–455.

Pushparaj, P., Kushwaha, B.P. and Prashar, S. (2025) 'A systematic literature review of virtual influencers in marketing using bibliometric analysis', *International Review on Public and Nonprofit Marketing*, Vol. 22, No. 3, pp.631–662.

Rao, A. and Tripathi, D. (2023) 'Credit accessibility and technological adoption in agriculture: evaluating government financial assistance programs in India', *Agrarian Studies Quarterly*, Vol. 12, No. 2, pp.156–172.

Reddy, S.S. and Kumar, D. (2023) 'Agricultural productivity and income growth in India: policy perspectives and empirical evidence', *Agricultural Economics Research Review*, Vol. 36, No. 2, pp.112–125.

Roy, T. and Pal, S. (2024) 'Inclusive agricultural digitalization: policy initiatives and farmer participation in India', *Technological Forecasting and Social Change*, Vol. 202, p.123829.

Rubel, Kushwaha, B.P., Sultana, R., Miah, M.H. and Sharma, S. (2025) 'An effective knowledge-based recommendation system for supply chain information based on UserCF algorithm', *International Journal of Business Innovation and Research*, Vol. 37, No. 2, pp.229–245.

Sarkar, S. and Sinha, R. (2022) 'Institutional support and income enhancement in developing economies: lessons from India's rural sector', *Economic Modelling*, Vol. 113, p.105916.

Sarstedt, M., Ringle, C.M. and Hair, J.F. (2017) 'Treating unobserved heterogeneity in PLS-SEM: a multi-method approach', in: Latan, H. and Noonan, R. (Eds.): *Partial Least Squares Path Modeling*, Springer, Cham, Heidelberg, Germany.

Sarstedt, M., Ringle, C.M., Smith, D., Reams, R. and Hair Jr, J.F. (2014) 'Partial least squares structural equation modeling (PLS-SEM): a useful tool for family business researchers', *Journal of Family Business Strategy*, Vol. 5, No. 1, pp.105–115.

Sen, A. (1999) *Development as Freedom*, Oxford University Press, New York.

Shah, N.P., Khakwani, A., Cinfio, F., Kushwaha, S. and Hu, X. (2026) 'Estimating the indirect economic burden of elevated lipoprotein (a): A 5-year simulation of cardiovascular events and associated patient and caregiver costs', *American Journal of Preventive Cardiology*, p.101562.

Sharma, N. and Meena, R. (2024) 'Farmer welfare schemes in India: an analytical review of income and empowerment impacts', *International Journal of Creative Research Thoughts*, Vol. 12, No. 4, pp.275–288.

Sharma, S. and Kushwaha, B.P. (2025) 'Sustainable consumption behaviour, values conscious purchase intention, and organic personal care products: extending customer values theory', *International Journal of Sustainable Society*, Vol. 17, No. 3, pp.282–301.

Singh, K. and Patel, N. (2023) 'Income equality and empowerment in Indian agriculture: examining the role of farm size and productivity', *Indian Journal of Social Sciences*, Vol. 18, No. 2, pp.55–70.

Singh, P. and Yadav, R. (2024) 'Financial inclusion and empowerment of indian farmers: the role of government programs and digital finance', *Journal of Agricultural Finance and Policy*, Vol. 36, No. 1, pp.66–83.

Singh, R. and Mehta, V. (2023) 'Government schemes and farmer empowerment: an analytical study of policy effectiveness in India', *Indian Policy and Governance Review*, Vol. 6, No. 1, pp.25–39.

Suresh, R. and Kaur, G. (2023) 'Equity in agricultural welfare programs: evidence from India's rural development schemes', *Policy and Society*, Vol. 42, No. 1, pp.89–107.

Suryavanshi, A., Baikar, A.K., Ruhil, M. and Kushwaha, B.P. (2025) 'Nexus of comprehensive growth and development parameters: a detailed analysis concerning BRICS countries', *Financial Innovation for Global Sustainability*, pp.141–159.

Vijayakumaran, N., Kushwaha, B.P. and Sakshi, S. (2025) 'Motivational factors driving franchise entrepreneurship: an empirical investigation', *International Journal of Business and Systems Research*, Vol. 19, No. 2, pp.203–220.

World Bank (2023) *Strengthening Agricultural Resilience through Asset Accumulation and Financial Inclusion In South Asia*, World Bank Publications, Washington, DC.

Zimmerman, M.A. (2000) 'Empowerment theory: psychological, organizational, and community levels of analysis', in Rappaport, J. and Seidman, E. (Eds.): *Handbook of Community Psychology*, pp.43–63, Springer, New York, USA.